

Revised ICLR Implementation Plan for ProActive

Executive summary

Your original document should remain the **execution backbone**. It already fixes the probe family, teacher-cache idea, budgeted acquisition, per-budget APS calibration, dataset suite, held-out shift tests, and repository/training scaffold. The professor's document should be treated as a **methodological override**: it changes the paper framing from "cheap diagnosis" to **active diagnostic measurement**, replaces the GRU as the main evidence encoder with **permutation-invariant set encoders**, adds **permutation-drift experiments** as a first-class result, strengthens baselines, and reopens the question of whether the six-way class should remain the scientific primary target.

The implementation-ready compromise that best merges both documents is this: keep the original **teacher, probes, budgets, state-sampling pipeline, and conformal reporting machinery**, but refactor the learning stack so that **Deep Sets is the default main encoder, masked-slot MLP is the cheapest invariant control, GRU stays as a serious baseline**, and **Set Transformer is optional unless probe interactions clearly matter**. This follows the professor's ranking of encoder roles and preserves the original project's execution realism. On targets, the cleanest merge is **dual-head training with asymmetric importance**: make the **three source bits** the main scientific head used for learning and analysis, keep the **six-way mapped label** as the human-readable reporting/calibration head, and explicitly say that the six-way output is a compressed presentation layer over richer behavioural evidence. That preserves your original six-way auditability while aligning the learning objective with the professor's source-bit-centric proposal.

Operationally, Weeks 1–4 from your original plan remain mostly intact. The real changes begin in Weeks 5–8: instead of a single GRU-centred stack, you now need a **shared-state interface** over four encoder conditions, a new **permutation evaluation script**, and a stricter **main-result hierarchy** centred on frontier quality, calibration efficiency, and robustness under permutation, model-family shift, and dataset shift. The mandatory additions are realistic within nine weeks if you treat Set Transformer, HALP, and BCEA-style baselines as **optional expansion**, not critical path.

What remains and what changes

The merged rule is simple: **keep the data-and-systems spine, replace the encoder-and-paper spine**.

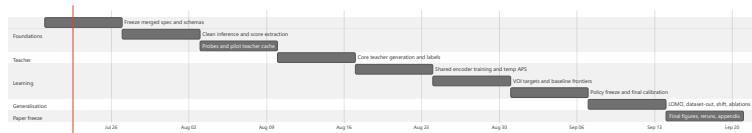
Original-plan section	Merged decision	Why
Project framing	Modify	Shift from “cheap HalluPrism-like diagnosis” to active diagnostic measurement under cost .
Probe family and STOP action	Keep	The original atomic probe set is already aligned with HalluPrism-style behavioural measurements.
Teacher cache	Keep	Full-evidence teacher cache is still the right substrate for offline VOI training.
Budgets and cost model	Keep	Budgeted acquisition remains central in both documents.
Dataset split logic	Keep	Grouped train/val/cal/test split still fits conformal calibration and leakage control.
Six-way target as sole primary target	Replace with dual-head design	Use source bits for primary learning; six-way mapping for reporting and APS calibration.
GRU as main encoder	Replace	GRU becomes a baseline; Deep Sets becomes the main model.
Policy learning via supervised counterfactual VOI	Keep	The professor strengthens this framing rather than replacing it with RL.
APS per budget	Keep, but clarify	Calibration must happen on the full frozen post-acquisition pipeline .
Existing baselines	Keep and expand	Add masked-slot MLP, GRU-as-formal-baseline, permutation-augmented variants, optional HALP/BCEA.
Evaluation	Expand	Add permutation drift, leave-one-dataset-out if feasible, and stronger transfer claims discipline.
Weekly plan Weeks 1–4	Mostly keep	These are data/teacher/foundation weeks and remain sound.
Weekly plan Weeks 5–9	Rewrite	Encoder family, permutation study, and dual-head calibration now change the downstream plan.

This mapping follows your original frozen scope on probes, datasets, budgets, and teacher construction, plus the professor’s explicit recommendation that masked MLP be the cheap invariant control, Deep Sets be the main model, Set Transformer be the interaction-aware upgrade, and GRU remain only the strong baseline/pilot.

The one design choice that should **not** remain frozen exactly as written in your original plan is the target hierarchy. Your document fixes the six-way label as the primary conformal object, whereas the professor’s design makes the source-bit evidence head central and treats the simplified label as optional reporting. The merged plan should therefore freeze the following hierarchy in Week 1: **primary training head = three source bits; auxiliary regression head = teacher signature; reporting/calibration head = six-way label**. This is the best balance between scientific fidelity and implementation risk.

Revised nine-week execution plan

The timeline below assumes the compute type, exact GPU count, and final sample caps remain **unspecified** in the supplied documents. The GPU-hour and storage estimates are therefore planning ranges, not claims of measured runtime.



The structure above keeps your original teacher-first pipeline but moves the professor's encoder logic into the centre of Weeks 5–8. That is the lowest-risk merge.

Week	Focus	Deliverables	Dependencies	GPU-hours	Storage delta	Go or no-go gate	Fallback action
Week 1	Freeze merged spec	Final target hierarchy; encoder interface; probe-token schema; split manifest; config schema	None	0–5	< 1 GB	One config can instantiate GRU, masked MLP, Deep Sets, Set Transformer	If target debate persists, freeze source bits as training head and six-way as reporting head
Week 2	Clean harness	Deterministic clean inference for all chosen models; score extraction; normalization tests	Week 1	15–30	2–4 GB	At least 2 models pass 50-example deterministic smoke suite	Defer the third model by one week; store only answer probability, entropy, margin, length
Week 3	Probes and pilot cache	All atomic probes; relation applicability mask; pilot teacher cache; probe sanity histograms	Week 2	25–45	2–5 GB	Probe outputs show nontrivial flips/shifts without saturation	Disable Set Transformer from critical path if probe signal is weak
Week 4	Core teacher and labels	Core teacher cache; source bits; six-way labels; state sampler v1; audit export	Week 3	80–140	8–15 GB	Two models fully cached; label histograms usable; no split leakage	Reduce sample caps on VizWiz and held-out data first
Week 5	Shared diagnostic training	Train clean-only MLP, GRU, masked MLP, Deep Sets; temporary APS; first permutation pilot	Week 4	30–55	4–8 GB	Deep Sets is at least competitive with GRU on validation frontier; invariant encoders show near-zero drift	Keep GRU and Deep Sets only; move Set Transformer out of schedule
Week 6	VOI and baselines	Realized VOI targets; random/fixed/uncertainty baselines; learned policies for GRU, masked MLP, Deep Sets	Week 5	25–45	3–6 GB	Deep Sets policy beats random and at least one fixed order at matched budget	Narrow to budgets {1,2,4}; postpone leave-one-dataset-out

Week	Focus	Deliverables	Dependencies	GPU-hours	Storage delta	Go or no-go gate	Fallback action
Week 7	Final policy and calibration	Freeze main stack; per-budget APS on six-way head; source-bit reporting tables; optional Set Transformer	Week 6	25–45	2–5 GB	Main Deep Sets stack beats GRU on either frontier quality or equal frontier with much lower drift	Keep Set Transformer appendix-only or cancel it entirely
Week 8	Transfer and robustness	Leave-one-model-out; optional leave-one-dataset-out; PRE-HAL/IllusionBench; ablations; latency	Week 7	45–90	3–6 GB	Learned policy remains above clean-only on LOMO; shift still shows useful empirical coverage	Reframe claim as within-distribution plus limited transfer
Week 9	Freeze and paper assets	Final figures/tables; seed reruns; appendix; risk memo; source pack	Week 8	10–25	1–2 GB	Reproducible frozen stack with no leakage	Cut optional baselines and keep only must-run tables

This revised plan preserves the original teacher-first build order, but it deliberately replaces the original “diagnostic GRU week” with a **shared encoder bake-off week**. That is the single most important schedule correction. The cumulative storage target should be planned as **35–60 GB free space**, because permutation logs and extra checkpoints will add modest overhead beyond the original JSONL caches. The total planning budget is roughly **255–480 GPU-hours**, with the teacher-cache stage still dominating.

Experiment matrix and implementation checklist

Experiment matrix

The matrix below is deliberately **minimal-but-reviewer-safe**. It avoids a full Cartesian explosion while still matching the professor’s methodological asks.

Condition	Encoder	Acquisition policy	Datasets	Metrics	Priority
Scalar confidence	None	No acquisition	Core + shift	AUROC incorrect-vs-correct, six-way set size baseline, coverage baseline	Must-run
Clean-only MLP	None	No acquisition	Core + shift	Source-bit micro/macro F1, six-way APS coverage/size	Must-run
Random	Shared diagnostic head	Random legal probe	Core	Cost, frontier, set size, coverage	Must-run
Fixed orders	Shared diagnostic head	Blank-first, visual-first, grounding-first, relation-first	Core	Cost–diagnostic frontier, per-dataset robustness	Must-run
Uncertainty-greedy	Shared diagnostic head	Myopic heuristic	Core	Frontier, stop behaviour	Must-run
GRU	GRU	Learned VOI	Core + LOMO + shift	Frontier, calibration, transfer, permutation drift	Must-run
Masked-slot MLP	Canonical-slot MLP	Learned VOI	Core + LOMO + shift	Frontier, calibration, transfer, permutation drift	Must-run
Deep Sets	Deep Sets	Learned VOI	Core + LOMO + shift	Main result: frontier, transfer, calibration, drift	Must-run
Set Transformer	Set Transformer	Learned VOI	Core + selected hard slices	Mixed-case gains, relation-heavy cases, drift	Optional unless clearly useful
Full teacher	Teacher	Full probes	Core + shift	Upper-bound quality, cost reference	Must-run
Oracle next action	Any	Oracle from cache	Core	One-step ceiling	Must-run
Oracle best subset	Any	Best subset up to budget	Core	Upper frontier bound	Must-run
HALP	One-pass probe on internals	No acquisition	One or two models only	AUROC, cost comparison	Optional
BCEA-style comparator	Verification-style	Answer/abstain/acquire	One claim-compatible slice	Coverage vs abstention	Optional

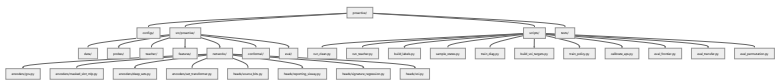
The central claim should be supported by four result clusters: **frontier quality**, **permutation stability**, **LOMO transfer**, and **calibration efficiency**. The professor’s memo makes clear that a mere in-distribution accuracy gain over GRU would be too weak.

Metrics that must be reported

Metric family	Required metrics	Meaning in this project
Diagnostic quality	Source-bit micro-F1, source-bit macro-F1, Hamming loss, six-way macro-F1	Whether the model diagnoses the right behavioural pattern
Calibrated sets	Coverage@90, Coverage@95, average set size, singleton rate	Whether the output is reliable and informative
Cost	Mean additional forward passes, stop rate, probe-count histogram	Whether the policy is actually budget-aware
Transfer	Leave-one-model-out, optional leave-one-dataset-out, IllusionBench, PRE-HAL	Whether the policy generalizes beyond the development slice
Permutation drift	Jensen–Shannon divergence, Jaccard disagreement, next-action agreement, hidden-state distance, calibration drift	Whether the encoder respects unordered evidence
Efficiency	Normalized latency vs clean pass, checkpoint size	Whether accuracy gains are practical

Deep Sets should win the argument if it maintains or improves frontier quality **while** collapsing permutation drift relative to GRU. That inference is strongly grounded in the theoretical fit of Deep Sets and Set Transformer to unordered evidence.

Repository refactor



Your original repository layout is already good enough; the necessary change is to split the old monolithic `encoders.py` into **encoder-specific modules** and to introduce a new `eval_permutation.py` path. The rest of the teacher/data/calibration scaffold can remain structurally similar.

Concrete implementation checklist

Area	Concrete decision	Immediate code impact
State representation	Replace ordered probe-prefix-only view with shared evidence-state interface	Add encoder-agnostic <code>EvidenceState</code> dataclass
Probe token schema	Use a single probe token format for all encoders	New tokenizer/featurizer reused by GRU, Deep Sets, Set Transformer
Heads	Train three heads: source bits, six-way reporting, teacher-signature regression	Split head modules; keep VOI head separate
Masked-slot MLP	Canonical 7 probe slots + acquisition bits	Slot tensor builder
Deep Sets	Per-probe token MLP + sum pooling + post-pooling MLP	<code>deep_sets.py</code>
Set Transformer	Small induced-attention block count only	<code>set_transformer.py</code> , but keep optional
GRU	Retain as baseline only	<code>gru.py</code>
VOI training	Keep offline realized-VOI training from teacher cache	Reuse <code>build_voi_targets.py</code> with encoder flag
Calibration	Calibrate only after full encoder + policy are frozen	Keep <code>calibrate_aps.py</code> , but feed post-acquisition states
Permutation study	New standalone evaluation path	<code>eval_permutation.py</code> and permutation log artifact
Transfer	Keep LOMO; add leave-one-dataset-out only if schedule allows	Extend <code>eval_transfer.py</code>
Logging	Log both bit-level and six-way predictions everywhere	Unified eval schema

Data caches and file contracts

The original teacher cache should stay JSONL-based. Do **not** store full logits by default; keep aggregate answer statistics and normalized answers only. That choice is already embedded in your original plan and remains right for this merged version.

Recommended cache stack:

Artifact	Purpose
<code>clean_*.jsonl</code>	Clean pass features and normalized answers
<code>teacher_*.jsonl</code>	Probe outcomes for all atomic probes
<code>labels_*.jsonl</code>	Source bits, six-way labels, teacher signature
<code>states_*.jsonl</code>	Partial states sampled for supervised training
<code>voi_*.jsonl</code>	Realized per-action VOI targets
<code>aps_thresholds.json</code>	Final per-budget thresholds for six-way reporting head
<code>perm_eval_*.jsonl</code>	Per-subset permutation outputs and drift metrics

Probe-token schema

A single probe event should be featurized as:

$$e_a = [\text{Emb}(a), \mathbf{1}_{\text{acquired}}, \mathbf{1}_{\text{applicable}}, \mathbf{1}_{\text{flip}}, \Delta c, m_{\text{exact}}, m_{\text{sem}}, \Delta H, \Delta m, c(a), \text{family}(a), \text{int}(g(x), o_a)].$$

In plain language, this means: every probe contributes a small packet saying **what probe it was, whether it was legal, whether it changed the answer, by how much confidence changed, whether the new answer matches semantically, how expensive the probe is, and how it interacts with the clean answer state**. This is exactly the sort of “behavioural evidence token” the professor is asking for, and it also aligns with the original cache fields you already planned to store.

Simplified concepts and LaTeX-ready equations

Why Deep Sets is the right default

A GRU assumes that the order of inputs matters. That is perfect for sentences or time series. Your probes are different: blank image, blur, crop, grounding, and relation swap are all **independent measurements on the same original example**. If the model has seen the same subset of probes, the diagnosis should ideally be the same no matter which one was observed first. That is exactly the inductive bias Deep Sets was designed for.

Use this as the main encoder equation:

$$h_t = \rho \left(\sum_{a \in S_t} \phi(e_a) \right), \\ z_t = [h_t; g(x); \text{Emb}(B_t)].$$

Here, (S_t) is the set of acquired probes by step (t), (e_a) is the token for one acquired probe, (ϕ) is a small per-probe embedding network, and (ρ) is a post-pooling network. The sum is what makes the representation permutation-invariant. If you swap the order of the same evidence tokens, the sum stays the same.

What value of information means here

Value of information is just a learned answer to one question: **“If I spend one more probe, how much more useful will my diagnosis become?”** Your original plan already operationalized this as a combination of **diagnostic-set shrinkage**, **diagnostic-loss reduction**, and **probe cost**; that structure should remain. $\square$

A clean implementation-friendly target is:

$$y_{\text{VOI}}(x, h_t, a) = \left(\left| \Gamma_{\alpha, B_t}^{\text{tmp}}(h_t) \right| - \left| \Gamma_{\alpha, B_t}^{\text{tmp}}(\hat{h}_t^a) \right| \right) \\ + \eta \left(L_{\text{diag}}(h_t) - L_{\text{diag}}(\hat{h}_t^a) \right) - \lambda_{B_t} c(a).$$

The first term says “does this probe make the calibrated set smaller?”, the second says “does it improve diagnosis even when set size does not move?”, and the third says “is the improvement worth the cost?” Your original plan already has the right logic; the professor mainly changes the state encoder feeding this target.

The one-step VOI regret bound

This is the modest theory claim worth keeping because it is true, readable, and useful.

Assume the true one-step value is $V(h, a)$, the estimated value is $\hat{V}(h, a)$, and the estimation error is uniformly bounded:

$$\left| \hat{V}(h, a) - V(h, a) \right| \leq \varepsilon \quad \text{for all legal } a.$$

Let

$$\begin{aligned} a^* &= \arg \max_a V(h, a), \\ \hat{a} &= \arg \max_a \hat{V}(h, a). \end{aligned}$$

Then the one-step regret is at most

$$V(h, a^*) - V(h, \hat{a}) \leq 2\varepsilon.$$

The proof is a two-line comparison: the chosen action can only be worse than optimal by the sum of the estimation errors on the optimal action and on the chosen action. This is not a grand theorem; it is exactly the kind of honest, modest proposition reviewers trust. It also matches the professor's preference for a simple solve-then-learn story rather than a complicated RL claim.

Why calibration must happen after acquisition

This is the calibration rule you should state very explicitly in the paper:

Freeze (π, θ) first, then calibrate on $s_{\text{final}}(x; \pi, \theta)$.

In words: once probe acquisition itself changes the model state and score distribution, you must calibrate the **final stopped pipeline**, not an earlier clean-pass score. This is exactly the lesson from adaptive conformal classification and recent budgeted evidence acquisition work.

For the six-way reporting head, APS remains a good first-choice method:

$$\begin{aligned} s_i &= \sum_{j=1}^{\rho_i} \pi_{(j)}(x_i), \\ q^\alpha &= \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(s_1, \dots, s_n). \\ \Gamma_\alpha(x) &= \left\{ y : \sum_{j: \pi_{(j)}(x) \geq \pi_y(x)} \pi_{(j)}(x) \leq q^\alpha \right\}. \end{aligned}$$

Keep APS in the mainline. Use RAPS only as a fallback appendix if sets are too large.

Resource-bounded selection and paper assets

Minimal resource-bounded configuration

If your adapters, prompts, and scoring code are already partly built around the original trio, the most sensible minimal configuration is:

Category	Recommended minimal set	Why
Models	Qwen3-VL-8B-Instruct, Gemma-4-E4B, InternVL3-9B	Minimizes replatforming cost while retaining architectural diversity
Core training datasets	HallusionBench, POPE or RePOPE, VizWiz-VQA, VSR, GQA relation slice	Matches both documents and gives object, illusion, poor-image, and relation coverage
Held-out shift datasets	PRE-HAL, IllusionBench	Best stress tests for reasoning-heavy and illusion-heavy transfer
Optional extra validation	AMBER	Cheap multidimensional secondary validation surface

This choice is conservative, but it is the right call if the project lead cares more about making the submission window than about re-anchoring every model family. Your original plan already uses the first three models, and the professor explicitly allows Qwen3-VL and Gemma as practical substitutions even while preferring Qwen2.5-VL, Molmo, InternVL3, and LLaVA-family models from a public-primary-source standpoint.

Expanded configuration

If you have budget for extra engineering, the expanded list should be:

Category	Expanded set	When to use it
Models	Qwen2.5-VL, Molmo, InternVL3 , plus LLaVA-OneVision as held-out family	Use if you are not locked into the original adapters or want stronger reviewer-facing external anchoring
Extra evaluation	AMBER and optional claim-style BCEA comparison slice	Use if frontier and calibration are already stable by Week 7

This expanded selection is more reviewer-friendly scientifically, because Qwen2.5-VL, Molmo, InternVL3, and LLaVA-OneVision all have strong primary-source support as modern open multimodal families. But it costs real time: new adapters, new scoring quirks, and more teacher-cache passes.

Recommended main figures and tables

The main paper should probably contain **five** central visual assets.

Asset	Main text or appendix	Why
Method overview diagram: image, probe controller, set encoder family, calibrated output	Main	Defines the paper’s object quickly
Cost–diagnostic frontier comparing clean-only, fixed, GRU, masked MLP, Deep Sets, full teacher	Main	This is the central empirical result
Permutation-drift figure across encoders	Main	This is the cleanest scientific validation of the encoder choice
Transfer table: LOMO, optional leave-one-dataset-out, PRE-HAL, IllusionBench	Main	Shows the architecture choice matters beyond the dev split
Calibration table: coverage and mean set size at 90% and 95%	Main	Makes the reliability story concrete

The appendix should absorb the rest: per-dataset breakdowns, per-budget tables, action-frequency plots, RAPS ablation, Set Transformer details, white-box HALP comparison if included, label-audit materials, and latency details. This split follows the professor's stronger framing around active measurement, invariance, and robustness rather than one-off ablations.

Risks, mitigations, and extra time

Risk	Why it matters	Mitigation	Extra time if triggered
Target ambiguity survives into training	Creates mismatched metrics and paper confusion	Freeze dual-head semantics in Week 1	2–3 days
Shared encoder refactor grows messy	Blocks Week 5	Create one evidence-state interface first, then plug encoders into it	3–4 days
Permutation study becomes too large	Eats evaluation time	Sample subsets and cap permutations at (P=10)	2–3 days
Set Transformer tuning churn	Risks blowing Week 7	Keep it optional until Deep Sets is already stable	3–5 days
White-box baselines expand scope	HALP/BCEA can easily drift out of control	Restrict to one or two models or move to appendix	4–7 days
Calibration leakage	Fatal for claims	Freeze encoder+policy before any final calibration pass	1–2 days
Teacher cache overruns	Still the dominant systems risk	Cap VizWiz first, then optional datasets, then fourth model	3–5 days

The practical estimate is:

- **Mandatory professor additions only:** roughly **+1.5 to 2 weeks** over the original GRU-centred plan.
- **Mandatory additions plus Set Transformer, HALP, and BCEA comparators:** roughly **+2.5 to 3.5 weeks**.

That is why the critical-path recommendation is: **Deep Sets main, masked MLP control, GRU baseline, Set Transformer optional, HALP/BCEA optional**.

Prioritized source list

The most useful source stack for writing and implementing this paper is:

Priority	Source	Use
Highest	Your original implementation plan	Execution backbone, probes, teacher cache, repo scaffold
Highest	Professor feedback memo	Encoder decision, framing, permutation study, stronger baselines
Highest	Deep Sets	Theoretical basis for unordered evidence encoding
Highest	Set Transformer	Interaction-aware invariant encoder
High	Classification with Valid and Adaptive Coverage	APS-style conformal classification
High	Uncertainty Sets for Image Classifiers using Conformal Prediction	RAPS fallback and set-efficiency framing
High	Look Again Before You Abstain	Post-acquisition calibration condition
High	HallusionBench	Core mixed-failure benchmark
High	POPE	Object-hallucination probing
High	VizWiz-VQA	Poor-image and answerability stress
High	VSR	Spatial-relation benchmark
High	GQA	Scene-graph-based relation slicing
High	PRE-HAL	Reasoning-heavy hallucination shift
High	IllusionBench	Illusion-heavy shift and trap cases
Medium	AMBER	Low-cost multidimensional validation
Medium	Qwen2.5-VL report	Strong modern anchor family
Medium	Molmo and PixMo	Fully open-data/open-weight family
Medium	InternVL3 report and model card	Architecturally distinct open model
Medium	HALP	Optional strong one-pass baseline
Medium	Qwen3-VL and Gemma official cards	Practical support for the original model stack

The final recommendation is therefore crisp: **do not rewrite the project from scratch**. Keep the original data, teacher, and systems plan; refactor the learning core around **Deep Sets + masked MLP + GRU baseline**; freeze a **dual-head target design** in Week 1; and make the paper stand on **frontier quality, permutation stability, calibration efficiency, and transfer**, not on a simple GRU-to-Deep-Sets swap.